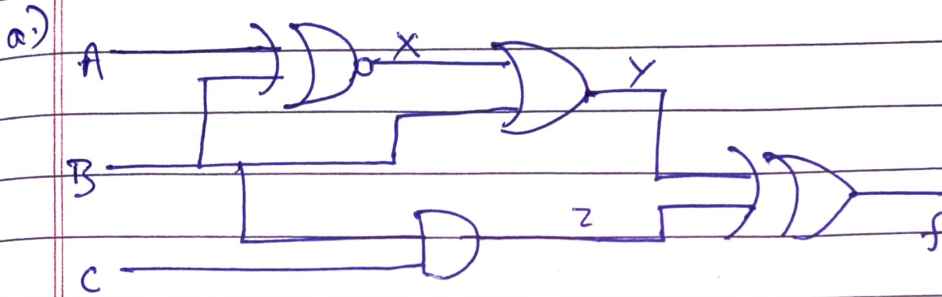


Day 2 - Assignment

1.) Derive the output expression 'f' for the below circuits and draw the simplified circuits.



$$A \oplus B' = AB + A'B' = X$$

$$Y = X + B$$

$$= AB + A'B' + B$$

$$Z = BC$$

$$Y = B + A'B'$$

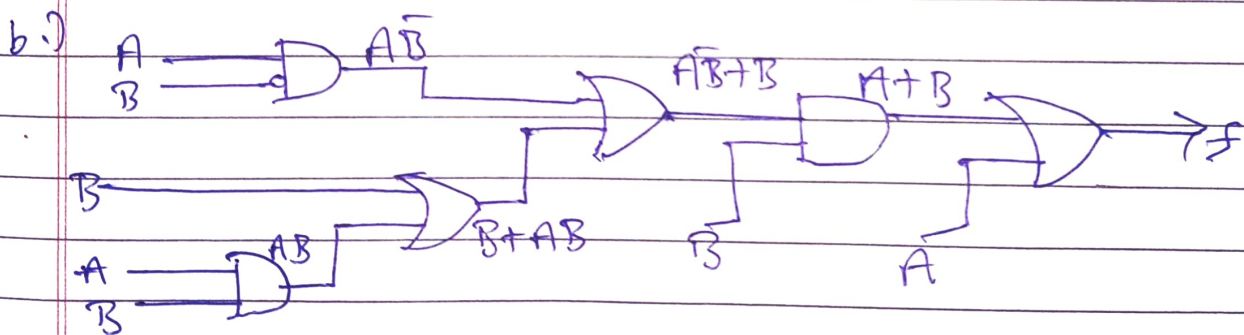
$$= B + A'B' = (B + A')(B + B')$$

$$= B + A'$$

$$Y = A' + B$$

$$f = (A' + B) \oplus BC$$

$$= A'B' + A'C' + BC'$$



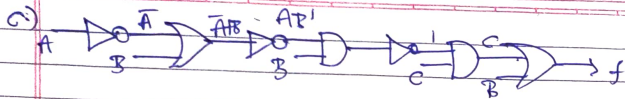
$$A\bar{B} + B + AB$$

$$A\bar{B} + B$$

$$B(A\bar{B} + B) = A + B$$

$$A + B + A$$

$$f \Rightarrow A + B$$



$$AB' \Rightarrow (AB')B = 0 = 1$$

$$f = C + B$$

2.) Minimise the following Boolean expressions using Boolean postulates.

a.) $\overline{A}CDE + ABD + ABCE + ABD$
 $= \overline{A}CDE + ABD + ABCE + ABD$
 $\Rightarrow CE(\overline{A}D + AB) + ABD + ABD$
 $= CE(AB + \overline{A}D) + ABD + ABD$
 $= ABCE + BDCE + \overline{A}DCE + ABD + ABD$
 $= ABC(E + D) + BDCE + \overline{A}DCE + ABD$
 $= ABD + BCED + \overline{A}BD$

b.) $\overline{A}B\overline{C} + ABC + \overline{A}C + \overline{A}C\overline{D} + A\overline{C}D + A\overline{B}C$
 $= \overline{A}B\overline{C} + ABD + \overline{A}C + A\overline{C}D + A\overline{B}C$
 $= \overline{B}C(\overline{A} + A) + ABD + \overline{A}C + A\overline{C}D$
 $= \overline{B}C + ABD + \overline{A}C + A\overline{C}D$
 $= \overline{B}C + \overline{A}C + AD(\overline{B} + \overline{C})$
 $= \overline{B}C + \overline{A}C + AD(\overline{C} + \overline{B})$
 $= \overline{B}C + \overline{A}C + AD(\overline{C} + \overline{B})$

c.) $\overline{A}C + BC + \overline{A}B + \overline{A}BC + A\overline{C}\overline{D} + \overline{B}C\overline{D}$
 $\overline{A}C + BC(\overline{A} + 1) + A\overline{C}\overline{D} + \overline{B}C\overline{D}$
 $\overline{A}C + BC + A\overline{C}\overline{D} + \overline{B}C\overline{D}$
 $(\overline{A} + \overline{A}C)\overline{D} + C(\overline{B} + \overline{B}C)\overline{D}$
 $\overline{A}C + BC + \overline{A}B$

d.) $\overline{x}_1\overline{x}_3\overline{x}_4 + \overline{x}_3\overline{x}_4 + \overline{x}_1\overline{x}_2\overline{x}_4 + \overline{x}_1\overline{x}_2\overline{x}_3\overline{x}_4 + \overline{x}_2$
 $\Rightarrow \overline{x}_1\overline{x}_3\overline{x}_4 + \overline{x}_3\overline{x}_4 + \overline{x}_1\overline{x}_2\overline{x}_3\overline{x}_4 + \overline{x}_1\overline{x}_2(\overline{x}_4 + \overline{x}_3)$
 $\overline{x}_4 + \overline{x}_1\overline{x}_4 + \overline{x}_3\overline{x}_4 + \overline{x}_1\overline{x}_2\overline{x}_3\overline{x}_4$
 $= \overline{x}_4 + \overline{x}_3\overline{x}_4 + \overline{x}_1\overline{x}_3$

Minimize the following Boolean expression using Kmap.

a.) $f(a, b, c, d) = \sum m(4, 6, 8, 10, 11, 12, 15) + \sum d(3, 5, 7, 9)$

	cd	cd	cd	cd
ab				
ab	1	x	x	1
ab	1		1	
ab	1	x	1	1

$$\overline{a}b + a\overline{b} + \overline{b}c\overline{d} + acd$$

b.) $f(a, b, c, d) = \sum m(1, 3, 5, 7, 12, 13, 14, 15)$
 $= \sum m(0, 2, 4, 6, 7, 9), \sum d(10, 11)$

	cd	cd	cd	cd
ab				
ab		1	1	
ab	1	1	1	1
ab				

$$\overline{a}b + \overline{a}d$$